



Refresh rate and motion

Most TVs run at 60Hz, but 120Hz panels handle fast motion more smoothly and reduce blur in sports and games.

Gaming features: VRR & ALLM

- **Variable Refresh Rate (VRR):** Matches TV refresh to console/PC frame rate, preventing screen tearing.
- **Auto Low Latency Mode (ALLM):** Automatically switches the TV to game mode for minimal input lag when a console is detected.

HDMI 2.1 essentials

HDMI 2.1 unlocks 4K@120 Hz, VRR, ALLM, and enhanced audio return channel (eARC) for lossless audio passthrough. Ensure your TV includes at least two HDMI 2.1 ports if you plan modern gaming or want to future-proof.

Smart platforms and user experience

Modern TVs serve as smart hubs – choose an operating system that fits your ecosystem and content habits.

- **Android TV/Google TV:** Best for deep Google ecosystem integration and wide app support.
- **Samsung Tizen:** Fast interface with Bixby and wide app catalog; ad-supported menus can feel cluttered.
- **LG webOS:** Clean UI with Magic Remote pointer; fewer third-party apps than Android but optimized performance.
- **Roku & Fire TV:** Simplified interfaces suitable for streaming-first users.

Case in Point: The Samsung S90D OLED combines Tizen's quick app launches with Alexa and Bixby voice assistants for hands-free control.

Connectivity and ports

A versatile TV must connect seamlessly to your devices:

- **HDMI:** Count and version. Aim for ≥ 2 HDMI 2.1 ports.
- **USB:** For firmware updates and media playback; USB 3.0 preferred.



- **Ethernet & Wi-Fi:** Gigabit Ethernet for stable streaming; Wi-Fi 6 for faster wireless.
 - **Audio Outputs:** eARC for AVR passthrough; optical SPDIF for soundbars; 3.5 mm audio jack for headphones.
- Pro Tip:** Label cables and leave slack for easier future swaps.

Budget and price ranges

Indian market prices (June 2025) roughly follow:

- **Budget (₹25,000–₹35,000):** 43"–50" Full HD or entry-level 4K.
- **Mid-Range (₹35,000–₹60,000):** 55"–65" 4K QLED or OLED alternatives.
- **Premium (₹60,000+):** 65"–75" OLED/8K MiniLED with advanced features.

Example: The Thomson Phoenix 55" 4K Google TV retails at ₹30,999 on Flipkart with Dolby Atmos sound.

Step-by-step TV selection checklist

1. **Measure & Calculate:** Determine viewing distance; apply $1.6\times$ diagonal for 4K or $1.5\text{--}2.5\times$ for HD.
2. **Choose Resolution:** Prioritise 4K for screens $\geq 55"$; consider 8K only for $\geq 75"$ seating scenarios.

